

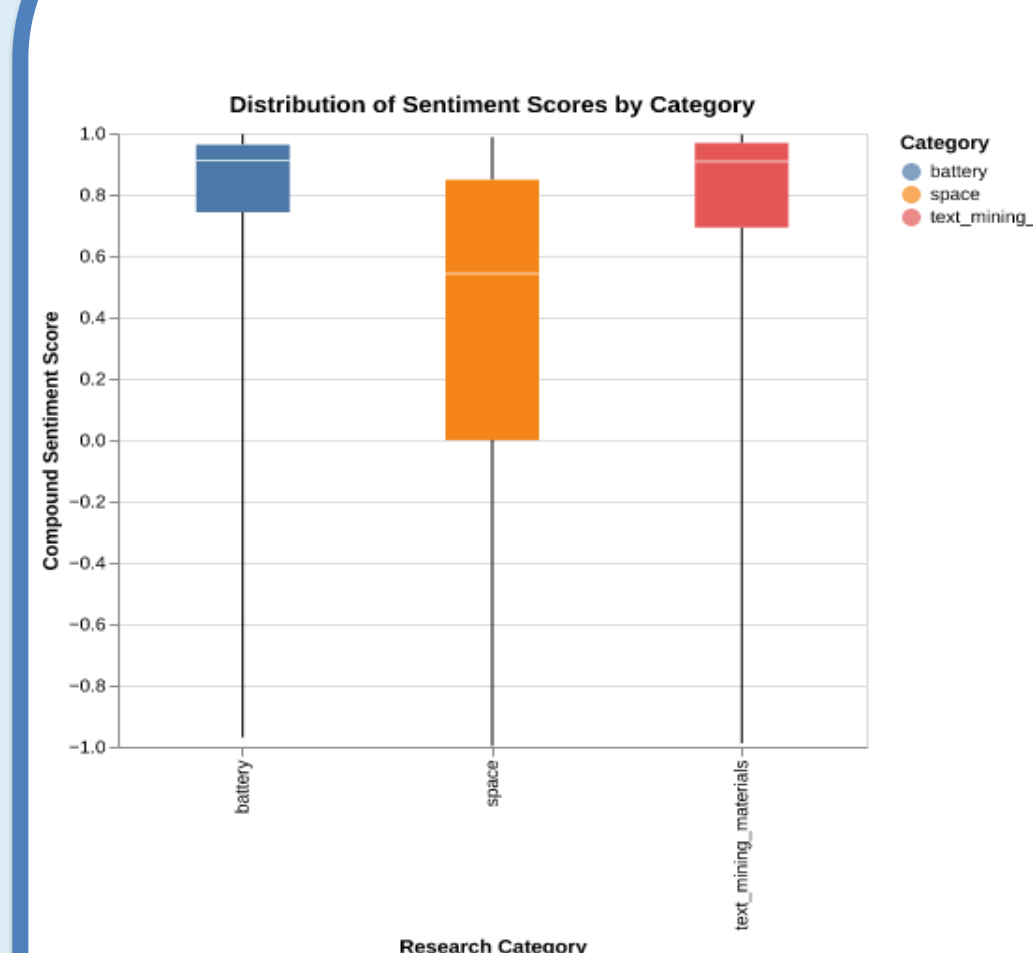
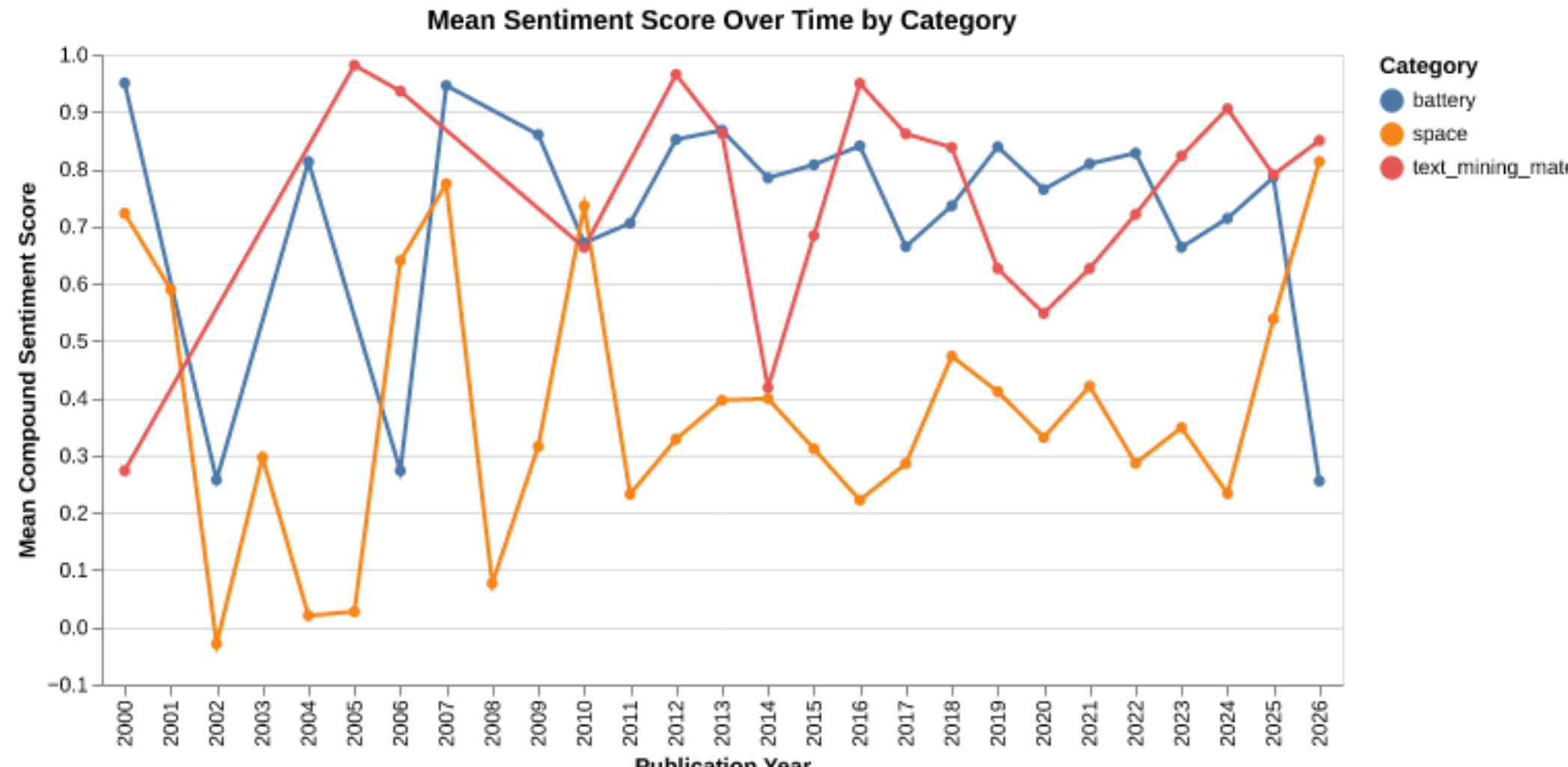
Discovering Patterns in Scientific Literature Abstracts using Text Mining and Machine Learning

Madison Bollinger and Nabin Joshi

University of Oklahoma

Introduction:

This project applies text mining and machine learning techniques to 1,400 scholarly abstracts collected from Lens.org across three research domains. Inspired by Tshitoyan et al. (2019), who demonstrated that scientific knowledge embedded in published abstracts can be extracted using NLP methods, this project performs a smaller-scale version of that work — discovering latent themes, comparing rhetorical tone, and building classifiers to predict research category from abstract text alone.



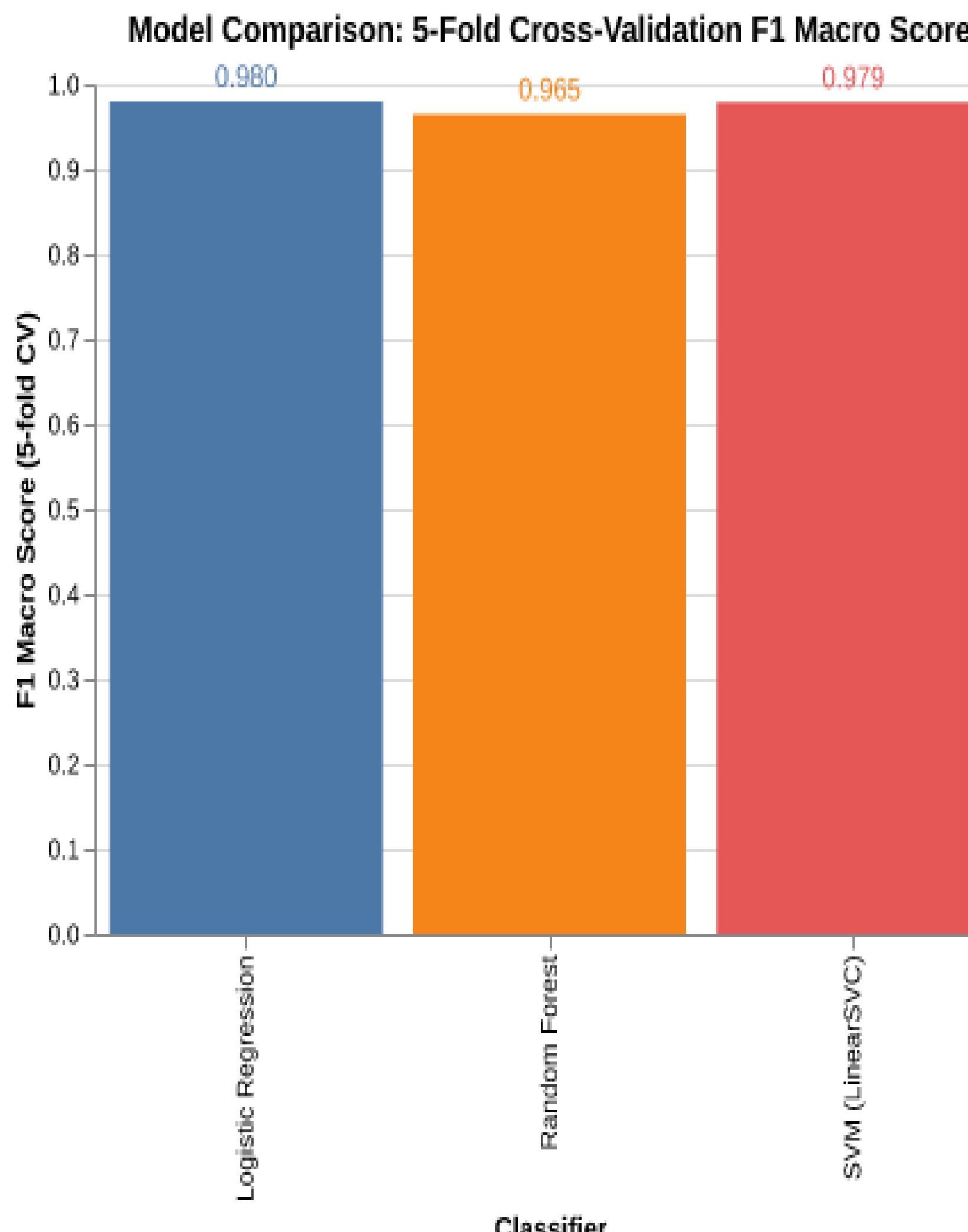
Data:

- ❖ Used Lens.org
- ❖ Created 3 searches: battery AND energy storage AND lithium, space AND satellite AND spacecraft, and text mining AND materials science AND battery
- ❖ Between the searches we have a total of 1400 rows of dataset



Methodology:

- ❖ Text Preprocessing Raw abstracts were cleaned through a 7-step pipeline.
 - ❖ Lowercasing, Number removal, Punctuation removal, Tokenization, Stopword removal (standard English + custom academic stopwords), Lemmatization (WordNetLemmatizer), and removed any blank abstracts (if applicable)
- ❖ The data we used is content heavy and we want to find the themes rather than the content itself, which can be provided by LDA.
- ❖ The models used include LDA, Logistic Regression, Random Forest, and Linear SVM.



Findings:

- ❖ The 10-topic LDA model (coherence = 0.4883) successfully uncovered distinct themes mapping closely to all three dataset categories without any supervision. This unsupervised discovery of the dataset's known structure validates both the preprocessing pipeline and the dataset quality — directly mirroring Tshitoyan et al.'s finding that scientific abstracts contain rich latent structure extractable without human labeling.
- ❖ Sentiment Analysis (VADER): 83.2% of all abstracts carry positive sentiment, with a mean compound score of 0.574 and a median of 0.827 — confirming that scientific writing is overwhelmingly optimistic in tone. Battery papers scored highest (0.751) followed closely by text mining materials (0.724), while space papers scored dramatically lower (0.351) due to their risk-aware, technically precise reporting culture.
- ❖ The dataset was converted into numeric vectors across 1,400 rows and split using an 80/20 train-test split. Logistic Regression, Random Forest, and SVM all reached 99% overall accuracy, with only a few misclassification errors across the models. Based on the 5-fold cross-validation macro F1 scores, Logistic Regression was the best model because it had the highest score at 0.980, narrowly outperforming SVM at 0.979 and Random Forest at 0.965.



Dataset in GitHub



GitHub Repository

Rational:

Text Analysis Method 1: Topic Modeling (LDA)

Tool: Gensim LDA + pyLDAvis

Since Tshitoyan et al. (2019) also used LDA approach — extracting latent thematic structure from scientific abstracts without supervision. It answers what themes exist across three research domains and is ideal for structured, content-rich scientific text.

Result: 10 topics mapped cleanly onto the three dataset categories without any labels provided.

Text Analysis Method 2: Sentiment Analysis (VADER)

Tool: NLTK VADER (SentimentIntensityAnalyzer)

Scientific abstracts frequently make confident, achievement-oriented claims. VADER reveals how researchers write across domains — comparing emotional tone as a complementary perspective to LDA's structural analysis. Network Analysis was not selected because it requires explicit

entity relationships not present at the abstract level.

Result: Battery papers (mean = 0.751) were significantly more positive than space papers (mean = 0.351); 83.2% of all abstracts are classified as positive.

Conclusion:

Overall, this project shows that text mining and machine learning can effectively identify meaningful patterns in scientific abstracts across battery, space, and text mining materials research. The LDA topic model successfully uncovered themes that matched the three dataset categories, while sentiment analysis showed that battery and text mining materials papers were generally more positive than space papers, which tended to use more risk-aware and technical language. The machine learning results were also very strong, with all three classifiers reaching about 99% accuracy; however, Logistic Regression was selected as the best model because it had the highest 5-fold cross-validation macro F1 score of 0.980 ± 0.011 . Space and battery papers were classified nearly perfectly because their vocabularies were distinct, while text mining materials was harder to classify because it shared vocabulary with both battery research and data science. Although the project was limited to three research domains and could be improved with expanded datasets, Word2Vec embeddings, or SciBERT-based sentiment analysis, the results demonstrate that text analysis methods can successfully classify scientific literature and reveal useful differences between research areas.

Reference:

Tshitoyan, V., Dagdelen, J., Weston, L., Dunn, A., Rong, Z., Kononova, O., ... & Jain, A. (2019). Unsupervised word embeddings capture latent knowledge from materials science literature. *Nature*, 571(7763), 95-98.